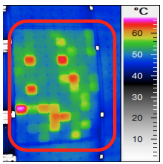
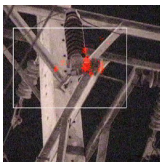
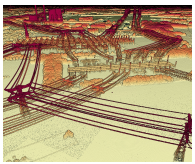
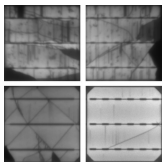
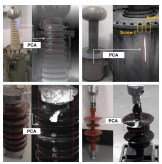
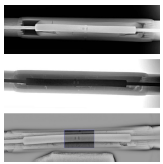
Principle	Passive			Active			
Band	Visible	IR	UV	IR		MSI / HSI	X-Ray
Wavelength	400-700 *nm*	8-14 μm	240-260 *nm*	980 - 1600 *nm*	1125-1175 *nm*	400 - 1700 *nm*	0.05 - 0.25 *nm*
Technology	Color Filtering	Thermal Irradiance Measurement	Corona Effect	Reflectance 3D Mapping	Luminescence Effect	Fluorescence Spectrum Analysis	X-Ray Penetration
Sensors	RGB Camera	IRT Camera	UVI Camera	LiDAR	InGaAs Camera Si CCD Camera	Hyperspectral Imaging Camera	X-Ray Scanner
Data Structure	2D × 3 Ch	2D × 1 Ch	2D × 1 Ch	3D × n[1] Ch	2D × 1 Ch	2D × n[2] Ch	2D × 1 Ch
Inspection Tasks	Surface anomalies Shape deviations Missing components Misplaced parts	Overheating components Faulty connections Insulation failures Load imbalances	HV discharges Insulation breakdown	Infrastructure profiling Tower tilting Cable sagging Vegetation encroachment	PV cell cracks Isolated defects Series resistance Thermal history	Oil leakage Vegetation detection	Internal defects Mechanical attachments and fittings of PL Conductors
Visual Samples							
Reference	[47]	[48]	[51]	[52]	[53]	[54]	[55]

[1] Point clouds may include channels like RGB, reflectance, or have no additional attributes.
[2] Multispectral typically has 4-10 channels, while hyperspectral can have tens to hundreds.